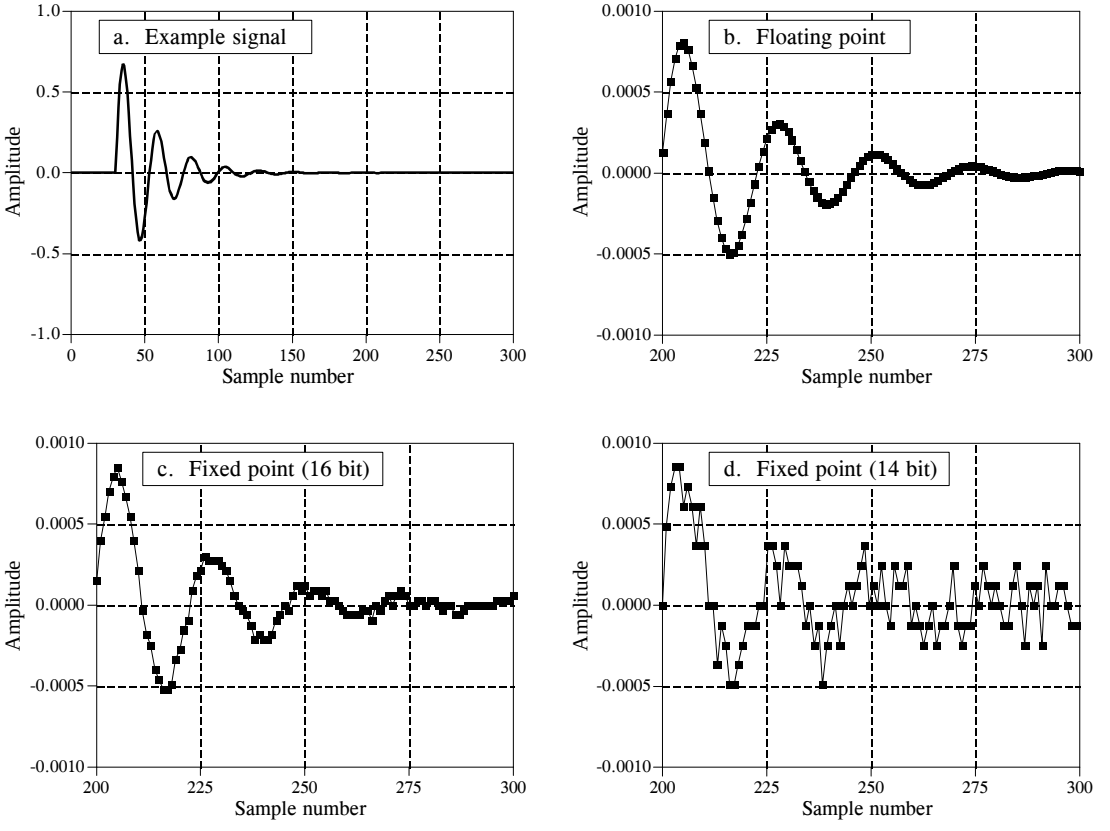

FIGURE 29-6
Round-off noise in the time domain. Figure (a) shows an example signal with an exponentially decaying amplitude. Figures (b), (c), and (d) show a more detailed view by zooming in on the amplitude. When the signal is represented in floating point, (b), the round-off noise is so low that it cannot be seen in this graph. As shown in (c) and (d), representing the signal in fixed point produces far higher levels of round-off noise.

filters are quite insensitive to round-off effects. As shown in Fig. 29-5, round-off noise doesn't change the overall shape of the frequency response; the entire curve just becomes noisier. IIR filters are a different story; round-off can cause all sorts of havoc, including making them unstable. Floating point allows these algorithms to achieve better performance in cutoff frequency sharpness, stopband attenuation, and step response overshoot.

Advanced Software Tools

Our custom filter example shows the *easiest* way to get a program running on the SHARC DSP: editing, assembling, linking, and downloading, performed by individual programs. This method is fine for simple tasks, but there are better software tools available for the advanced programmer. Let's look at what is available for when you get really serious about DSPs.

The first tool we want to examine is the **C compiler**. As discussed in the last chapter, both assembly and C are commonly used to program